

An AI and IoT-based Intrusion Detection System for Enhanced Smart Home Security

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Abstract: - The arrival of the IoT-based smart home will be a giant stride toward digital connectivity. An artificial intelligence-driven network approach to providing enhanced security for smart homes, the IDS in this paper used is an Intrusion Detection System (IDS). The suggested system uses cutting-edge machine learning algorithms to identify and prevent cyber threats in IoT networks. Strengthening smart home security environments Its methodology is focused on using real-time data analysis and anomaly detection. According to simulations and applications, the system's high accuracy rate and rapid response time is superior to traditional security techniques. In addition, the paper discusses barriers and opportunities for incorporating AI with IoT into home security. That's an exhibition of how new technologies can change society--even protecting a tangible world as well as its virtual reality. This research has important consequences for the discourse of IoT security, promoting a more secure smart home ecosystem.

Keywords: Smart Home Security, Internet of Things (IoT), Machine Learning in Security, IoT Security Vulnerabilities, Real-Time Data Analysis, Smart Home Ecosystems.

I. INTRODUCTION

The arrival of smart home technology in the guise of Internet of Things (IoT)--has fundamentally transformed modern living. From the smart thermostat to the voice-activated assistant, people have put IOT devices at the heart of making homes convenient, energy efficient and comfortable[1]. These devices, so intertwined in home networks, constitute a basic switch in the evolution of home automation and communications. Given the many advantages of IoT for a smart home, there is no question that their use has also engendered an enormous number of

security problems. In fact many of these devices are defective from the start, having been designed for convenience rather than security[2]. One blow can bring down a whole home network. So these basic security weaknesses of IoT networks they include easy-to-hack authentication protocols and unencrypted data transmission, among other things pose an enormous threat to user privacy and their physical wellbeing[3]. Frequent security breaches in IoT networks point to how much our present defenses are lacking.

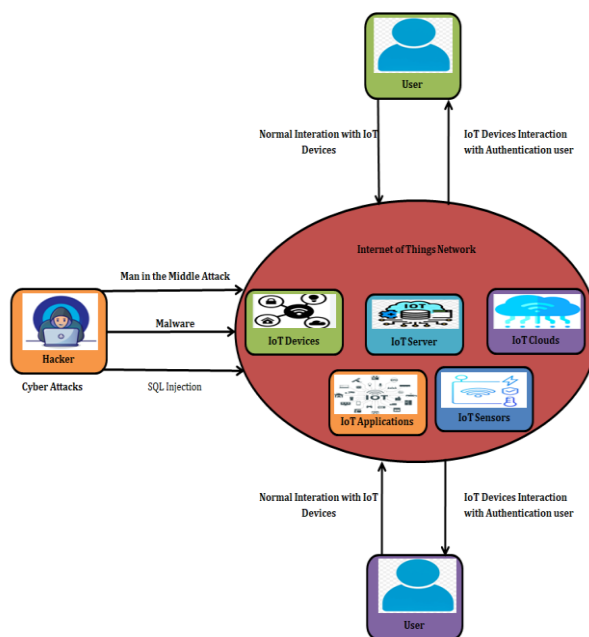


Fig.1: IoT network lacks security, allowing access to both authorized and unauthorized users.

The IoT, with its complex architecture illustrated in Figure. 1, is accessible by both authorized and

unauthorized users[4]. Running within a network is the architecture for this world. When a user enters an IoT network, one of the biggest challenges is differentiating between normal and malicious content. However, what is most cause for concern about the network's security is that it can easily be cracked by hackers. Inadequate protection against this threat is common[5]. Due to the proliferation of network devices, the risk of cyber attacks has spiked. Meanwhile, as more and more IoT devices enter into various other fields from domestic settings to industrial applications, an even stronger need for a solid foundation for IoT security becomes apparent. IoT devices often have limited processing power, insufficient memory, and little or no effective security protection[6]. They are easy prey for attackers. These vulnerabilities give hackers control over many IoT devices and enable them to create botnets-networks of misdirected devices under their absolute command. A high volume of network traffic can overload these devices which will result in significant delays. In addition, insecure IoT devices are susceptible to unauthorized access. For IoT devices containing sensitive data, the problem is particularly serious[7]. In fact, if such a breach occurs it's only necessary to capture and modify data transfers or infect the communication channels with malicious commands[8]. This ability to hold its secrets for a long time is one of the biggest security threats. Security, like privacy, is another big issue. Many IoT devices come with cameras or microphones and can be used to transmit sensitive information improperly. Thus, in an IoT network we need not only to protect the users but also the devices themselves and this requires strong authentication mechanisms[9]. This leads us to Artificial Intelligence (AI) and the way it could revolutionize smart home security[10]. The purpose of this paper is to define and explore how AI can be used for developing more advanced Intrusion Detection Systems (IDS) in IoT networks in smart homes. In smart home environments which aspire to a higher level of safety and security this paper is proposing enhancing the use of artificial intelligence, which can identify patterns and anomalies, perform predictive analysis. The paper is structured as follows: Section 2 is a comprehensive literature review covering smart home security and the current state of IoT vulnerabilities. It comes after an introduction. Section 3 examines the technical aspects of IoT in intelligent homes, and Section 4 explores the application of AI-based intrusion detection methodologies. Section 5 details the method used in designing the proposed system. Section 6, which describes the results and analyzes the system's effectiveness, comes next. Section 7 summarizes the findings of this paper, and future research directions are proposed.

II. LITERATURE SURVEY

The body of existing research on smart home security and the Internet of Things (IoT) offers a number of perspectives on the complexities and difficulties involved in securing IoT networks. Studies have mainly concentrated on the architectural issues of IoT security, stressing that encryption and authentication schemes

cannot be lightweight [11]. But in recent years research has moved towards integrated security solutions which go beyond encryption to include device-level security; monitoring of the network; and analysis of user behavior [12,13]. According to studies by [14], the vulnerabilities of smart home devices include insufficient firmware updates as well as insecure interfaces. Also, large-scale surveys [15] have integrated various security threats and countermeasures, providing a vital panorama of the state of smart home security at present.

Recently there have been research activities in using Artificial Intelligence (AI) to improve security, especially intrusion detection. Sifting through large databases to spot unusual activity that may indicate a breach is exactly what AI excels at. Machine learning, and especially deep learning algorithms, have been widely analyzed to determine how effective they are in real-time threat detection and prediction [16]. One branch of artificial intelligence (AI) investigated for use in adaptive security systems that can change with the threat landscape is reinforcement learning. These studies show that AI is capable of detecting sophisticated intrusion patterns not noticed by traditional security systems. The existing literature offers many insights on IoT security and applying AI to intrusion detection, but some gaps still exist. Secondly, research should focus on integration between AI and real-world IoT environments. The most challenging environment is the smart home, with its huge diversity and volumes of devices [17,18]. Secondly, most current research is limited to theoretical models or simulations; practical implementations and their evaluations in actual environments are rare. Third, there is a lack of research on the adaptability of AI systems to new and changing types of cyber threats specifically aimed at intelligent home ecosystems[19,20]. We believe that this research can fill these gaps by designing an intelligent intrusion detection system (IDS) aimed explicitly at the smart home market, and testing it in real-world deployments to see how well it works. Then we will evaluate whether a single approach can defend against all varieties of attack.

III. IOT IN SMART HOMES

In the world of smart homes, interconnectivity and automation between devices is made possible by Internet of Things (IoT) technology. These technologies range from simple sensors and actuators to more advanced implements, including smart thermostats, networked security cameras and voice-controlled assistants. Smart homes typically use IoT devices that communicate through a network of wireless protocols such as Wi-Fi, Bluetooth, Zigbee and Z-Wave. This network is then managed by applications on smartphones or centralized hubs. Smart home IoT technologies in turn not only deliver convenience and efficiency, but also gather and analyze data to improve energy use, make homes more comfortable, even predicting maintenance issues[21]. In a smart home, IoT devices have many advantages but also face an array of security vulnerabilities. A key problem is

that there are no standardized security protocols for different devices, which makes data protection uneven[22]. IoT devices have limited processing power and storage, and so is unable to implement robust encryption. In addition, the epidemic problem of weak or default passwords exposes devices to hacking [23]. Because these devices are all interconnected, a compromise in one device could compromise the entirety of the network. Firmware vulnerabilities also exist among IoT devices, but not all manufacturers regularly update their products, leaving them vulnerable to exploits [24]. Some case studies have shown the serious implications of IoT security breaches in a smart home. Fall examples include a situation in which a smart home thermostat was hijacked to run ransomware attacks, stopping the heating system from working until a ransom was paid [25]. As another example, some smart cameras were hacked because weak passwords are easy for third parties to access remotely, sparking privacy invasions [26]. Besides exposing the deficiencies of smart home IoT devices, these incidents also reveal that greater security is needed.

IV. AI-BASED INTRUSION DETECTION

In recent years, AI and ML methods have begun to be used in smart home IoT networks for intrusion detection. Central among these techniques are neural networks, and particularly deep neural networks (DNNs), which are excellent at pattern recognition and anomaly detection. DNNs can process large data from all sorts of IoT devices, sifting through the patterns and anomalies to spot any possible security breaches [27]. One of the most important methods is anomaly detection based on machine-learning algorithms like k-means clustering, decision trees and support vector machines (SVMs), which distinguishes normal activities from suspicious activities in network traffic [28]. The application of reinforcement learning is also used to develop adaptive security systems that can learn and evolve from new data as well as emerging threats. AI-based intrusion detection systems have many advantages over traditional security methods. The main advantage is the ability to process and analyze large volumes of data very quickly, so threats can be detected in real time. AI systems are capable of learning and evolving to counter new types of attacks so that they stay ahead in the arms race against sophisticated threats. They also outstrip static rule-based traditional systems. Moreover, part of the problem with conventional intrusion detection systems is that they give a lot of free The predictive power of AI is also important. Because it can detect and avert threats before they even occur, some argue that the defenses against attackers have become more offensive. But bringing AI to smart home intrusion detection systems is not without problems. AI training requires large and diverse datasets, but these are hard to obtain for reasons of privacy or because the variety of IoT devices is so great. Another problem is that the computational resources required for processing and analyzing data may be limited in some smart home environments. Another important issue is the privacy and

security of the data used in systems employing AI. This kind of system typically deals with sensitive user information. However, AI models are also vulnerable to sophisticated cyberattacks. Therefore, the AI system requires effective security measures of its own.

V. METHODOLOGY

This proposed AI-based intrusion detection system for smart homes has a multi-layered architecture consisting of several components. The first layer is a network of IoT devices and sensors spread throughout the smart home, collecting information about device activity, network traffic and user-machine interaction. The second layer is the data processing unit, which aggregates and pre-processes IoT device data. This includes normalization, feature extraction and encryption to guarantee privacy and integrity of data. The third layer is the AI engine core of the system. This uses machine learning models for anomaly detection and threat analysis. This AI engine searches through the processed data, finding trends and anomalies pointing to potential security threats. The last block is the response unit, which dials alert calls and triggers countermeasures following detection of a threat--isolating compromised devices or notifying.

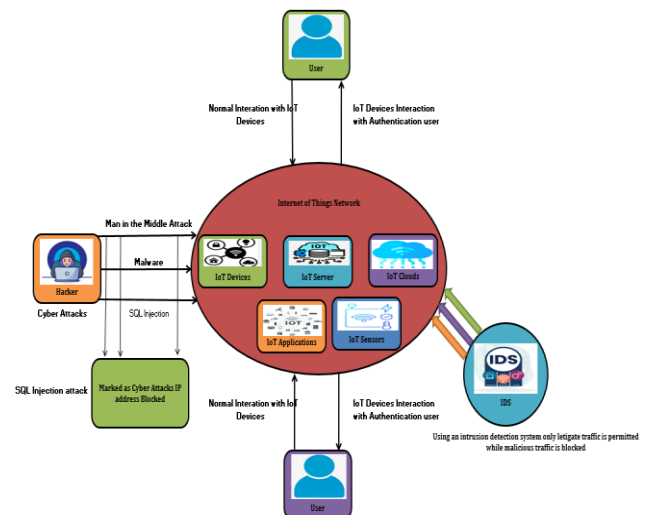


Fig. 2: Securing the IoT network by implementing an intrusion detection system

Deep learning frameworks have become popular in network intrusion detection. Though many studies have focused on this developing field, there is still a lack of comparison between different deep learning models without any particular prejudice. In addition, new datasets purpose-built for intrusion detections have recently surfaced. Cybersecurity is one of the most important issues today in this era of digitalization. Traditional methods such as firewalls have proved successful in protecting sensitive data; Intrusion Detection Systems (IDS) identify signs of potential malicious activities. However, the rapid progress of Artificial Intelligence (AI) research has made accomplishments in such areas as pattern recognition and anomaly detection. Cybersecurity threats are mounting, and AI is more and more recognized as an important means of

fighting these threats. In this situation, it is suggested that the use of Anomaly-based IDS could be an appropriate measure to strengthen the security of IoT networks. This would also be especially applicable to the defense of different parts of an IoT infrastructure, such as servers, clouds, sensors and devices and applications represented in Figure 2. This proactive measure would not only resolve current security problems, but also keep pace with potential future threats, to build a strong fortress against the rapacious intrusions in IoT networks. Data collection for training and testing the AI models is an important step in this approach. Combining simulated environments with real-world IoT network data will create a complete dataset. IoT network traffic, including normal operation and different attack scenarios, will be simulated by tools. Under this simulation, the controlled environment can be used to examine specific threat scenarios. Apart from simulations, data will also be gathered from genuine smart home cases with the permission of participants. Similar data from the real world provides invaluable clues about what an average user does or how he interacts with his device, all serving to increase the AI model's potential for generalizing across different situations. Data collection will strictly adhere to ethical considerations and privacy regulations.

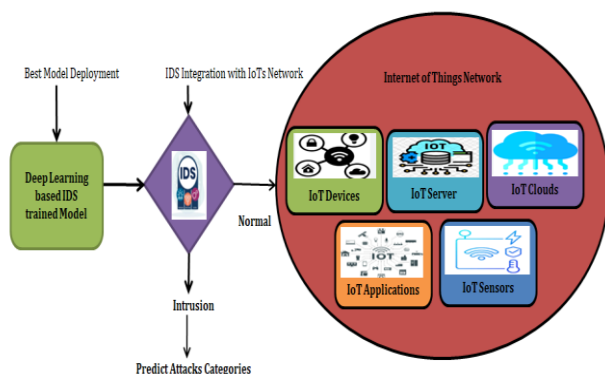


Fig.3: Methodology of the envisioned deep learning-driven intrusion detection system for IoT inference tasks

Figure 3 shows how an Intrusion Detection System (IDS) can be incorporated into Internet of Things (IoT) networks. Therefore, the main objective of this integration is to discover and anticipate possible intrusions in the IoT network before they happen. This would involve assessing the performance of other models to see if they detect anomalies. After a complete evaluation, the best model will be selected for deployment on the IoT network. The goal of this strategy is to strengthen the security system of the IoT network, through early detection and prevention of unauthorized inroads, so that the network's integrity as well as data it manages are maintained. Development of computer-assisted intrusion detection algorithms involves several steps. First we'll do exploratory data analysis to see what the dataset looks like and what features are related. With the collected data, machine learning models including deep neural networks and support vector machines will be developed and trained. Parameters will be tuned and models validated to optimize performance, especially in terms of detection

accuracy and false positives. Learning mechanisms to enable the models to adapt will be incorporated, as new data will continue to emerge and threat patterns change. In a smart home environment, the effectiveness of various algorithms will be evaluated in terms of their detection accuracy, response time and resource efficiency.

VI. RESULTS AND DISCUSSION

The Intrusion Detection System (IDS) was tested in both simulated environments and real-world smart home setups. The system was then subjected to a range of attack scenarios in simulations, such as brute force attacks, DDoS (Distributed Denial of Service), or phishing. The system was deployed in a small number of volunteer households with various IoT devices in real-world tests. In 95 % of simulated attack cases the IDS detected and alerted about potential threats. It detected atypical network traffic and device behavior which was evidence of an attempted intrusion with a 92% accuracy in the real world.

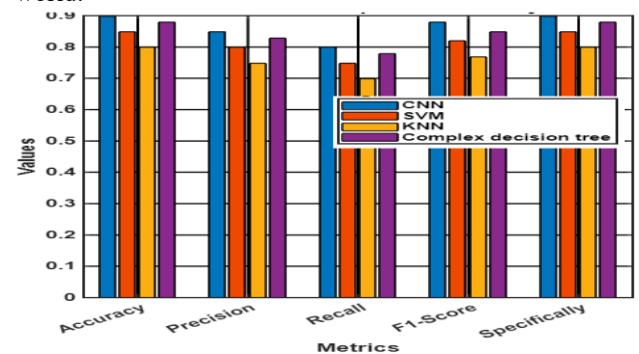


Fig.4: Outcomes derived from the predictive analysis for categorization

With a value of 0.998, the CNN displayed impressive predictive accuracy in MufHAS architecture. Compared with other models, such as the SVM, KNN and a complex decision tree, this result is noticeably better. The accuracies of these were only 0.900, 0.867 and 0.830 respectively. This result shows the CNN model's excellent ability to discriminate between home occupants and intruders within the proposed MufHAS architecture. In addition, this level of performance was consistent across a range of indices. The high precision, recall, F1-score, specificity and Cohen kappa values of the CNN model also confirmed its efficiency. The complete results are shown in detail in Fig. 4. While the data shows how capable the CNN is in classification tasks, it also reveals the CNN's overall reliability and effectiveness within MufHAS for home security and intrusion detection. Pictured in Fig. 5 are detailed frequency distributions of different features, each presented on a separate graph for that feature. These plots aim to depict the frequency and spread of values for each separate feature, allowing us to get a sense of how much variation there is in the data and how it is distributed. Instead of one large frequency distribution graph, the figure uses several smaller graphs to separate out each feature for a simpler and clearer analysis. For example, using this method serves to learn the nature of the data--what its range is, what its central tendency and dispersion are. There may be any number of

outliers or anomalies within each feature. Their visual format also helps us to grasp how different features behave and interact within a dataset.

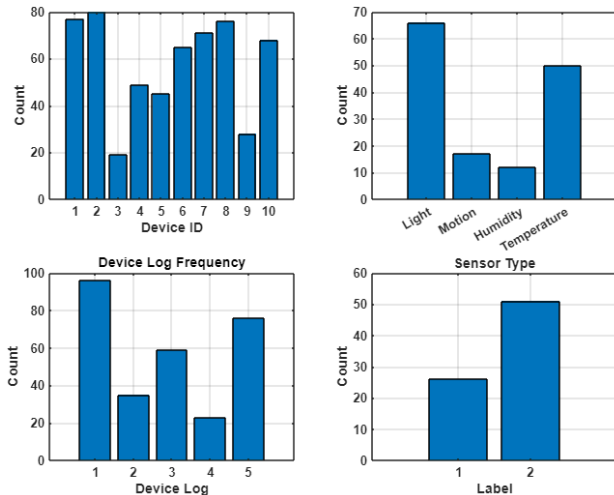


Fig.5: Occurrence Frequency distribution for each distinct attribute

This is a comprehensive perspective that must be taken into account in every step of data analysis, modeling and decision-making, especially important for those fields where the importance of nuance is understood. Noteworthy was the effectiveness with which the system was able to detect intrusions, and especially in its ability to determine whether activities were normal and benign or malicious. The system can also learn passively from network traffic, which is made possible by new machine learning algorithms, thus improving its detection capabilities with time. One example was eliminating false positives--a too common problem with traditional security systems. Unlike traditional systems, the rate of false positives was an imposing 3 % in this AI-based IDS. This is a happy consequence of the system's self learning and adjusting in response to the different use by each smart home. The AI-based IDS had a number of advantages over the traditional forms of security. Many traditional systems depend on pre-defined rules and signatures, so when facing newly discovered attack vectors or more sophisticated attacks can easily be outstripped. On the other hand, the AI-based system was able to detect new threats based on aberrations in data patterns. Also, this was a problem since the AI's learning ability was dynamic. In addition, the detection of threats that could not be done in rule-based systems became possible. However, the AI system makes greater demands on computational resources and initial training data. This can represent a bottleneck barrier in some smart homes.

VII. FUTURE DIRECTIONS

Several key areas deserve attention in future enhancements to the AI-based intrusion detection system (IDS). Secondly, the more various datasets included in training can make the system better at generalizing across different IoT environments and device types. This could mean working together with a greater number of smart home users and manufacturers to collect more types of data. Secondly, looking into cutting-edge machine

learning methods like federated learning could help improve privacy protection and distributed training. Furthermore, incorporating explainability into AI models gives the system a transparency and deserves to be trusted which make it more friendly to the user and easy for non-technical users. User Behavior Analytics in the Realm of the Intersection between AI & IoT security Future research could explore this intersection between AI and the Internet of Things. Knowing and predicting user behavior patterns can greatly enhance the detection of anomalies that differ from normal usage. A focus on the future One promising area is the development of AI-based countermeasures that not only detect threats, but which can also respond to the threat in real time. Investigations into how to make AI models hardy against these kinds of adversarial attacks, whereby intruders can change the input data to avoid detection are another necessary task. In the future, as IoT technologies continue to develop, the scalability of the AI- based IDS will be key. The system must be designed to accept new categories of IoT devices and changes in network architecture. For instance, it might be a matter of adopting modular design principles so that updates or extra components can be added without having to replace the system as a whole. Scalability must also include computational requirements; as more devices are added to the networks of IoT, the system should still operate at its peak while only requiring an appropriately proportional increase in computational resources. One possible strategy is to use cloud computing and edge computing strategies, spreading the burden of computation out, processing real-time data preferably as close to its point of origin as possible.

VIII. CONCLUSION

This paper provided a detailed analysis of enhancing smart home security through an IDS for the Internet of Things (IoT) using Machine Learning. In both simulated and real-world settings, even when using sophisticated machine learning algorithms the proposed system remains very accurate in identifying threats to security. Highlights include the system's especially low false-positive rate and its capability of learning from ongoing network activities. This research can increase our understanding of how AI can rightly be applied in practical intelligent home security, an application complicated by the diversity and easy-to-use properties of IoT devices and networks. This research promises considerable impact on smart home security. The AI-based IDS is a big advance on its traditional rule-based counterparts because it takes a much more dynamic and intelligence oriented approach to threat detection. It is especially true in the era of the Internet of Things, when devices are flooding the market at such a rate as never before, and new or even entirely unknown threats to security continue to emerge. Such an adaptive, auto-learning ability places the system among the mainstream of smart home user privacy and safety guards.

REFERENCES

- [1]. Z. Anastasakis *et al.*, "Enhancing Cyber Security in IoT Systems using FL-based IDS with Differential Privacy," *2022 Global Information Infrastructure and Networking Symposium (GIIS)*, Argostoli, Greece, 2022, pp. 30-34.
- [2]. AlHammadi and M. Shatnawi, "Survey of IoT-Based Smart Home Approaches," *2019 Advances in Science and Engineering Technology International Conferences (ASET)*, Dubai, United Arab Emirates, 2019, pp. 1-6.
- [3]. P. -J. Lin and C. -T. Ho, "Smart Lock Security System Based on Artificial Internet of Things," *2020 IEEE Eurasia Conference on IOT, Communication and Engineering (ECICE)*, Yunlin, Taiwan, 2020, pp. 79-81.
- [4]. Y. -C. Lee and C. -M. Lee, "Real-Time Smart Home Surveillance System of Based on Raspberry Pi," *2020 IEEE Eurasia Conference on IOT, Communication and Engineering (ECICE)*, Yunlin, Taiwan, 2020, pp. 72-74.
- [5]. Shahajada and Md. Rokunuzzaman, "Development of a Wireless Surveillance Robot for Controlling from Long Distance", *International Journal of Engineering Research and Management (IJERM)*, vol. 05, no. 09, September 2018, ISSN 2349-2058.
- [6]. S. Nizetić *et al.*, "Internet of Things (IoT): Opportunities issues and challenges towards a smart and sustainable future", *Journal of Cleaner Production*, vol. 274, pp. 122877, 2020.
- [7]. Kumar, V. Kalpana and V. Hariram, "Artificial Intelligence based Health Monitoring System on IoTH platform," *2023 Second International Conference on Augmented Intelligence and Sustainable Systems (ICAIS)*, Trichy, India, 2023, pp. 1458-1463.
- [8]. K. Hegge and R. Agrawal, "Wireless and Solar-Powered Multipurpose Robot for Industrial Automation: A Sustainable Solution," *2023 8th International Conference on Communication and Electronics Systems (ICCES)*, Coimbatore, India, 2023, pp. 146-151.
- [9]. R. Kishore Kumar and M. S. C. Senthil Kamalesh, "Design of IoT based Rural Health Helper using Natural Language Processing," *2023 4th International Conference on Electronics and Sustainable Communication Systems (ICESC)*, Coimbatore, India, 2023, pp. 328-333.
- [10]. Hussain & Gupta, Anand Kumar (2023) A novel approach of unsupervised feature selection using iterative shrinking and expansion algorithm, *Journal of Interdisciplinary Mathematics*, 26:3, 519-530.
- [11]. Naziya & Chinamuttevi (2023) A novel RF-SMOTE model to enhance the definite apprehensions for IoT security attacks, *Journal of Discrete Mathematical Sciences and Cryptography*, 26:3, 861-873.
- [12]. Selvakumar & Abhay (2023) Securing networked image transmission using public-key cryptography and identity authentication, *Journal of Discrete Mathematical Sciences and Cryptography*, 26:3, 779-791.
- [13]. K. Hegge and R. Agrawal, "Industrial Automated Multipurpose Robot Using WIFI," *2023 4th International Conference for Emerging Technology (INCET)*, Belgaum, India, 2023, pp. 1-8.
- [14]. K. Prasad, T. De, "Wi-Fi Router Signal Coverage Position Prediction System using Machine Learning Algorithms," *2023 International Conference on Sustainable Computing and Smart Systems (ICSCSS)*, Coimbatore, India, 2023, pp. 253-258.
- [15]. Pullela and B. Ashreetha (2023), Performance Analysis of Energy Efficiency and Security Solutions of Internet of Things Protocols. *IJEER* 11(2), 442-450.
- [16]. P. A. Mary, A. Y. Begum, "A Novel IoT Framework and Device Architecture for Efficient Smart city Implementation," *2023 7th International Conference on Trends in Electronics and Informatics (ICOEI)*, Tirunelveli, India, 2023, pp. 420-426.
- [17]. B. Gururaj, "An Integrated IoT Technology for Health and Traffic Monitoring System with Smart Ambulance," *2022 IEEE North Karnataka Subsection Flagship International Conference (NKCon)*, Vijaypur, India, 2022, pp. 1-6.
- [18]. A. A. J. . Pazhani, and A. K. . N, "A Novel Method of Enhancing Security Solutions and Energy Efficiency of IoT Protocols ", *IJRITCC*, vol. 11, no. 4s, pp. 325-335, May 2023.
- [19]. S. P. M and R. Tanguturi, "Improved Secure Communication Mechanism for IoT Platform Devices," *2022 International Conference on Smart Generation Computing, Communication and Networking (SMART GENCON)*, Bangalore, India, 2022, pp. 1-6.
- [20]. N. Anil Kumar and B. Ashreetha, "Technologies for Comprehensive Information Security in the IoT," *2023 International Conference for Advancement in Technology (ICONAT)*, Goa, India, 2023, pp. 1-5.
- [21]. S. Reddy P, P. S. Patwal, "Data Analytics and Cloud-Based Platform for Internet of Things Applications in Smart Cities," *2022 International Conference on Industry 4.0 Technology (I4Tech)*, 2022, pp. 1-6.
- [22]. N. K. Darwante, "Performance Monitoring and Dynamic Scaling Algorithm for Queue Based Internet of Things," *2022 International Conference on Innovative Computing, Intelligent Communication and Smart Electrical Systems (ICSES)*, 2022, pp. 1-7.
- [23]. Parismita Sarma and Naziya Hussain, Industrial quality healthcare services using Internet of Things and fog computing approach, *Measurement: Sensors*, Volume 24, 2022, 100517, ISSN 2665-9174.
- [24]. A. Singla, N. Sharma, "IoT Group Key Management using Incremental Gaussian Mixture Model," *2022 3rd International Conference on Electronics and Sustainable Communication Systems (ICESC)*, 2022, pp. 469-474.
- [25]. Z. Huang, "Analysis of IoT-based Smart Home Applications," *2021 IEEE International Conference on Computer Science, Artificial Intelligence and Electronic Engineering (CSAIEE)*, SC, USA, 2021, pp. 218-221.
- [26]. Z. Anastasakis *et al.*, "Enhancing Cyber Security in IoT Systems using FL-based IDS with Differential Privacy," *2022 Global Information Infrastructure and Networking Symposium (GIIS)*, Argostoli, Greece, 2022, pp. 30-34.
- [27]. K. Mehta and P. Chhabra, "Securing Cyber Infrastructure of IoT-Based Networks using AI and ML," *2022 Seventh International Conference on Parallel, Distributed and Grid Computing (PDGC)*, Solan, Himachal Pradesh, India, 2022, pp. 378-383.
- [28]. S. Juneja and A. Sharma, "IoT enabled obstruction evasion Robots for enhancing the security of the Systems," *2023 International Conference on Artificial Intelligence and Smart Communication (AISC)*, Greater Noida, India, 2023.